Simulation-Based Framework for Autonomous Navigation of Vehicles in Controlled Environments

Alby Jacob Martin
Department of Computer Science and Engineering
Alliance School of Advance Computing
Alliance University, Bengaluru, India
jalbyBTECH23@ced.alliance.edu.in

Yeshwanth R
Department of Computer Science and Engineering
Alliance School of Advance Computing
Alliance University, Bengaluru, India
ryeshwanthBTECH23@ced.alliance.edu.in

Ritam Jana
Department of Computer Science and Engineering
Alliance School of Advance Computing
Alliance University, Bengaluru, India
jritamBTECH23@ced.alliance.edu.in

Anupama K Ingale
Department of Computer Science and Engineering
Alliance School of Advance Computing
Alliance University, Bengaluru, India
anupama.ingale@alliance.edu.in

***Abstract*—In this paper, we propose a simulation-based retrofit approach for autonomous navigation in vehicles operating in controlled environments. Instead of removing existing vehicles, this approach focuses on integrating additional hardware which helps with autonomous navigation in controlled environments. This approach uses graph-based environment representation and the A* algorithm for calculating optimal paths. A dashboard-based interface is developed to visualize navigation, monitor system status, and dynamic replanning. To support hardware integration, a circuit design is developed and validated using simulation tools, representing sensor input, control logic, and actuation components. The overall system is implemented as a proof-of-concept and evaluated through simulation tests. The results demonstrate that the approach can calculate optimal paths, adapt to environmental changes, and provide a foundation for the deployment of real world autonomous vehicles in controlled environments.**

***Index Terms*—Autonomous Navigation, A* Algorithm, Path Planning, Controlled Environment, Dynamic Replanning, Simulation-Based Validation, Dashboard Monitoring**

I. Introduction

Autonomous navigation systems have received considerable attention and interest in recent times due to the potential benefits of improved efficiency and safety in the transportation and logistics domains. Controlled environments such as warehouses, industries, university campuses and shipyards are more suitable for the implementation of autonomous vehicles due to the highly structured and organized nature of infrastructure and traffic patterns [7], [8]. Although significant improvements and innovations are being made in the field of autonomous driving systems, the adoption of these systems is still in its infancy due to the high costs of the hardware components required for the system's implementation. Most existing systems are designed and implemented under the assumption of a completely autonomous platform and are thus difficult to implement in industries and organizations with existing manually driven vehicles [1], [4]. In this regard, a software-centric approach to autonomous navigation systems can be a viable alternative to the traditional methods of proving the feasibility of the system before moving on to the actual hardware implementation. Classical path planning algorithms such as the A* algorithm are known to perform efficiently in structured environments due to the deterministic nature of the algorithm and the optimality of the paths obtained [6], [7]. This approach can thus be a viable alternative to the traditional methods of proving the feasibility of the system before moving on to the actual hardware implementation. This paper propose a proof-of-concept (PoC) software framework for the implementation of autonomous navigation systems in controlled environments. This paper aims to prove the feasibility of the system and thus provides a platform for the actual hardware implementation of the system in the future. The system aims to integrate the perception abstraction mechanism with the grid-based environment modelling and the A* algorithm to generate the required paths for the autonomous system to navigate while avoiding obstacles and finding the shortest paths possible. The system is implemented and simulated within a simulated environment with the aid of a visualization tool to monitor the behaviour of the system in real-time and thus assess the efficiency of the system in different scenarios. The key contribution of this work is the design and simulation-based validation of a lightweight, modular autonomous navigation framework that leverages classical path planning techniques within a grid-abstracted environment.

II. Literature Review

Current research on this topic has grown significantly in recent years, considering its importance in both academic and practical contexts. Earlier researches have explored different methods and techniques, contributing to our knowledge base along with some constraints. Some of the relevant literatures are cited here, which are among the factors that have inspired us for this work.

A. Path planning algorithms

One of the significant parts of autonomous navigation is the path-planning algorithm, which helps the vehicle find the way from its starting point to the destination by avoiding any obstacles present in between [7], [8]. Classical path planning algorithms like Dijkstra and A* are commonly used because of deterministic and optimal behaviour. Among these, A* has been shown to perform particularly well in structured and grid-based environments, because of its low computational cost. An optimized version of the A* algorithm has been proposed by Chu et al. [6] for intelligent vehicle path planning, showing improvements in path efficiency and planning time through heuristic functions. This shows the applicability and importance of classic path planning algorithms like A*, it is further highlighted by Zhu et al. [10], who combined this algorithm with vision based intelligent vehicles, which allowed it to navigate through combination of classic path planning and vision based perception. Path planning based on sampling such as RRT and RRT*, are also commonly used, especially when dealing with complex and unstructured environment. An improved version of RRT* has been proposed by Li et al. [9] and combined with an artificial potential field method to improve path planning quality and accuracy. Even though these path planning algorithms perform effectively, they also introduce uncertainty, creating a loss in determinism and predictability. This has been highlighted by Reda et al. [7] and Shi [8] in survey based studies, showing how classical path planning algorithms like A* are still the best choice for path planning in controlled environments because of their ease of use and interpretability.

B. Autonomous Navigation in Controlled Environments

This suggests autonomous navigation in structured or semi-structured areas like factories, campuses, ports, and warehouses is quite different from autonomous navigation on roads due to structured paths, traffic rules, and low-speed navigation. There have been various studies on autonomous driving in structured or semi-structured areas, where navigation paths and rules have been considered pre-defined and motion rules have been set. In a study conducted by Wang et al. [1], a real-time trajectory planning method for autonomous vehicles was proposed based on pre-defined reference paths and cost functions. In another study, Meng et al. [3] proposed a digital twin based testing environment for autonomous vehicles to test and verify vehicle behaviour in virtual environments. In this study, physical vehicles have been connected to virtual platforms for efficient testing and validation. In another study, Yang et al. [4] proposed an autonomous driving approach based on an overall system view, where trust, security, and cooperation have been considered for Internet of Vehicles. In another study, Majid and Arshad [5] proposed an autonomous navigation approach for surface vehicles based on swarming systems. In this study, it can be seen that autonomous navigation for surface vehicles is possible in structured areas. However, this study is based on a domain-specific environment where autonomous navigation for ground vehicles based on retrofitting is not possible. From all the above studies, it can be seen that autonomous navigation in structured areas is possible, but it is based on complex systems, high-quality maps, or domain-specific assumptions.

C. Sensor-Based Perception for Navigation

In the case of sensor-based perception, the importance is clearly shown in the role that sensors play in the autonomous navigation system, where they are used to detect obstacles and make appropriate decisions. The use of camera-based sensors, as an example, has been extensively studied and shown to be cost-effective and provide sufficient information regarding the surroundings. Zhu et al. [10] presented the application of camera-based object detection in the implementation of the A* path planning algorithm, indicating the importance that the output of the perception system plays in the decision-making process. Range-based sensors, as an example, are often used to detect obstacles and estimate the distance, and they are accurate in their results. However, most autonomous systems assume the availability of expensive sensors and high computational power, and Sung et al. [2] indicated the importance of sensor data collection and analysis, although the application was focused on data logging rather than actual autonomous navigation. The current state-of-the-art in the implementation of the autonomous system, as indicated by the importance that the output of the perception system plays, is focused on the accurate and complete sensing of the surroundings, and this is often achieved through the use of computationally expensive techniques. However, in the case of controlled environments, the opportunity exists to implement the autonomous system using simplified perception abstractions.

III. Research Gap

Existing literature on autonomous vehicle navigation systems has clearly proven its efficacy in developing sophisticated path planning techniques, perception-based navigation systems, and large-scale validation systems. In this regard, various graph-based techniques like A* and optimization-based path planning techniques have been effectively utilized for developing efficient navigation systems under various operating conditions [6], [7]. However, it is observed that existing techniques have been primarily developed for large-scale autonomous vehicle systems operating in highly dynamic or public domain conditions. These systems have been primarily designed to operate with high-end sensor systems, computational systems, and integration complexities, which may not be suitable for cost-sensitive or hardware-based systems. Moreover, existing literature has not addressed software-based techniques that emphasize determinism, computational efficiency, and ease of deployment for operating in controlled conditions. Additionally, existing literature has primarily addressed hardware based validation techniques, while relatively less emphasis is given to software-based proof-of-concept techniques for evaluating navigation systems. This brings us to the proposed work, where an attempt is made to develop a modular-based autonomous vehicle navigation system using a simulation based approach. By utilizing classical path planning

techniques and perception-based abstraction techniques, an effective and reliable basis is established for future hardware-based implementations.

IV. Methodology

A. System Overview

The proposed framework follows a modular and simulation driven architecture for autonomous navigation in controlled environments. The overall framework is divided into four primary modules: perception, environment modelling, path planning, and navigation control. The system works in a graph-based controlled environment, where paths and obstacles are defined to simulate real-world conditions such as campuses or industrial layout. A* path planning algorithms is used to compute optimal path between a given start and goal position. The perception layer is represented through a simulation test, where obstacle information is given dynamically to reflect environmental changes. A dashboard-based interface is developed to visualize the environment, monitor system behaviour, and interact with navigation inputs in real-time. The modules have been constructed in such a way that they can work autonomously but with a constant flow of data in the entire system. This will help to facilitate the verification of the navigation algorithms without any hardware components, through simulation. [1]–[3], [6], [7], [10]

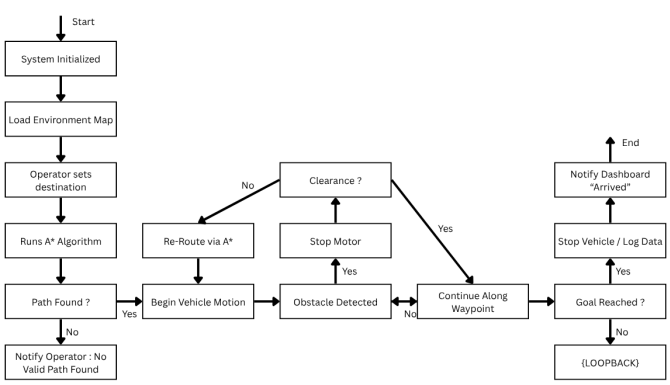

Fig. 1. Activity Diagram of Dashboard Interface

B. Graph based Environment Modelling

Controlled environment of the moving vehicle is assumed to be 2D grid, where each grid cell corresponds to discrete physical area. Grid cells can be classified either as navigable or non-navigable, depending on the environment characteristics and perception module inputs. Static obstacles such as trees, walls are defined during map initialization, while dynamic obstacles are determined through real time updates. This grid based representation gives a controlled and computationally efficient system for path planning, which allows path planning algorithms to be applied directly.

Each grid cell is treated as a node is search space, with adjacency defined by neighbouring cells. Movement costs between adjacent cells are assigned based on distance and travel conditions. The grid resolution is selected to balance path accuracy and computational load, ensuring real time feasibility within controlled environments. In terms of implementation efficiency, the grid based environment has been represented as graph, where certain grid cell locations are represented as nodes, and possible transitions between these locations are represented as edges. This representation has allowed the path planning problem to be solved through graph traversal, while still maintaining consistency with the graph based approach. [6], [9], [10]

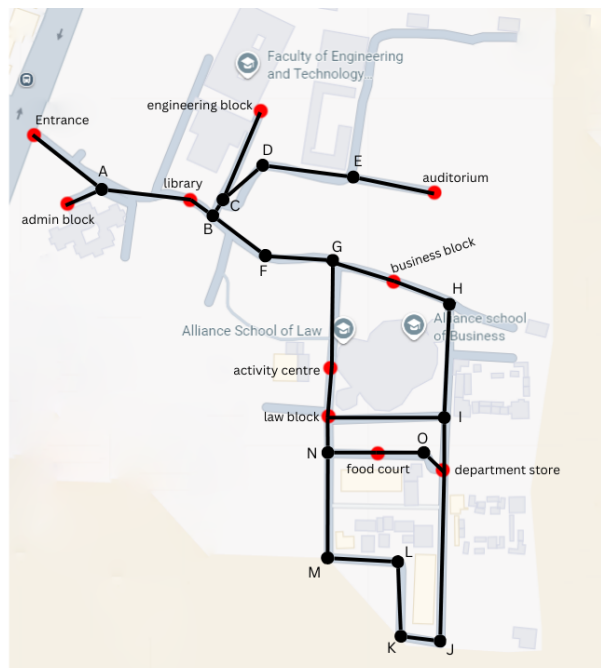

Fig. 2. Custom map used in dashboard for testing

C. Perception and Obstacle Representation

Sensor inputs are processed by perception module, focusing on obstacle detection and localization rather than raw sensor management. Simulation based perception techniques are used for identifying static and dynamic obstacles within the environment. Detected obstacles are then shown in the environment model to reflect the changes in real time. This method ensures that the perception module is separated from the planning logic, allowing flexibility in adapting to different sensing modals while maintaining consistent module inputs. [3], [9], [10]

D. Path planning using A*

Path planning is performed using the A* algorithm due to its optimality and efficiency in controlled environments. The module calculates an optimal path from a defined start cell S to a goal cell G minimizing an evaluation function defined as:

$$f(n) = g(n) + h(n) \tag{1}$$

where g(n) represents the cumulative cost from the start cell to the current cell n, and h(n) is a heuristic estimate of

the cost from n to the goal. In this work, the heuristic function is based on the Euclidean or Manhattan distance between grid cells, depending on allowed movement constraints. The A* algorithm operates on the graph representation derived from the grid-based environment, and nodes are expanded in a manner such that they are ordered based on an increase in cost. This helps to avoid non-navigable nodes, which represent obstacles in the path, ensuring a path that is optimal and free from obstacles based on the environment provided. This algorithm has a low computational cost, and its determinism makes it suitable for use in a controlled environment. [6], [10]

Algorithm 1 A* Path Planning Algorithm

Require: Grid map M, start node S, goal node G
Ensure: Optimal path from S to G

```
Initialize open list and closed list
Add start node S to open list
Set g(S) = 0, compute f(S) = g(S) + h(S)
while open list is not empty do
    Select node n with lowest f(n) from open list
    if n is goal node then
        Reconstruct path from S to G
        return path
    end if
    Move n from open list to closed list
    for each neighbor m of n do
        if m is not traversable OR in closed list then
            continue
        end if
        Compute tentative cost:
        g_temp = g(n) + cost(n, m)
        if m not in open list OR g_temp < g(m) then
            Update parent of m as n
            g(m) = g_temp
            f(m) = g(m) + h(m)
            if m not in open list then
                Add m to open list
            end if
        end if
    end for
end while
return failure (no path found)
```

E. Path validation and Dynamic Replanning

To ensure safe navigation in the presence of dynamic obstacles, the generated path is continuously validated against updated perception data. If an obstacle is detected along the planned route, the affected grid cells are marked as non-navigable, triggering partial or full re-planning. This allows the system to adapt to the environmental changes while taking care of deterministic behaviour and avoiding unnecessary recalculation. [4], [9], [14], [17]

F. Simulation and Software Level Validation

The proposed system has been validated using a simulation-based POC framework, allowing systematic validation of its path planning performance without depending on any other hardware implementation. A virtual graph based environment is developed, allowing dynamic definition of obstacles, start and goal position. Performance metrics include path length, computation time and re-planning frequency. These simulations provides mathematical information on the information on the performance of the system and validates the feasibility of the proposed methods before integrating to real world. [3], [12], [17], [20]

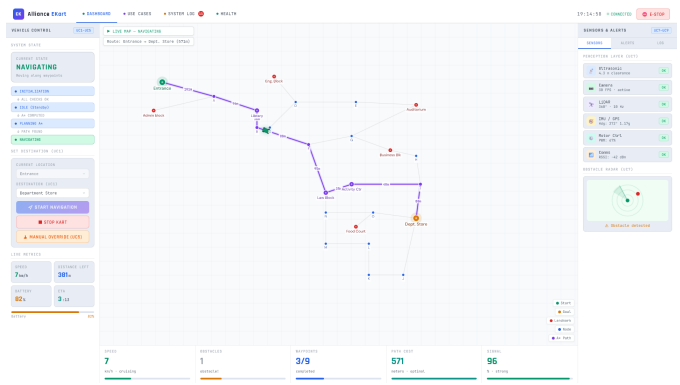

Fig. 3. Dashboard Interface

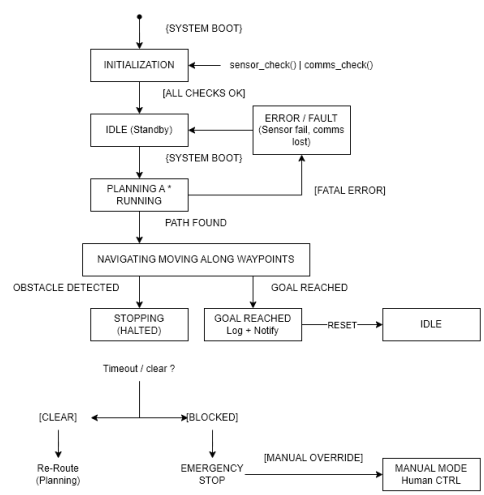

Fig. 4. State Diagram

G. Proof of Concept Implementation

The proposed framework is demonstrated through a software based proof of concept system, where environment modelling, path planning, and visualization methods are integrated. The system allows user to set start and goal position, configure obstacles and view the resulting path for navigation calculated by A* path planning algorithm. A dashboard based interface is developed to visualize the graph environment, identify obstacles and display the path for navigation in real time. The concept is demonstrates the feasibility of integrating classical

path planning along with lightweight perception in software based system. [3], [6], [9], [10], [14], [20]

H. Hardware Design and Circuit Simulation

The proposed system is not only autonomous navigation, but hardware layer is also required for real-world deployment. Hardware implementation for this work is part of future work where we plan to integrate this system with real sized vehicle on a controlled environment. We have designed and validated the hardware architecture using circuit simulation. In our circuit simulation, we have used components such as Arduino Uno R3 as controller, various sensors for different inputs, and actuator for output. The sensors are used for detecting obstacles, that data is then sent to controller. Controller processes logic and sends signal to motors(movement) and servo(steering). LCD is attached to hardware system to check system status. As mentioned earlier, we have not implemented this using real hardware, but using simulation to validate connections before real implementation, this approach reduces cost and complexity. [2]–[4]

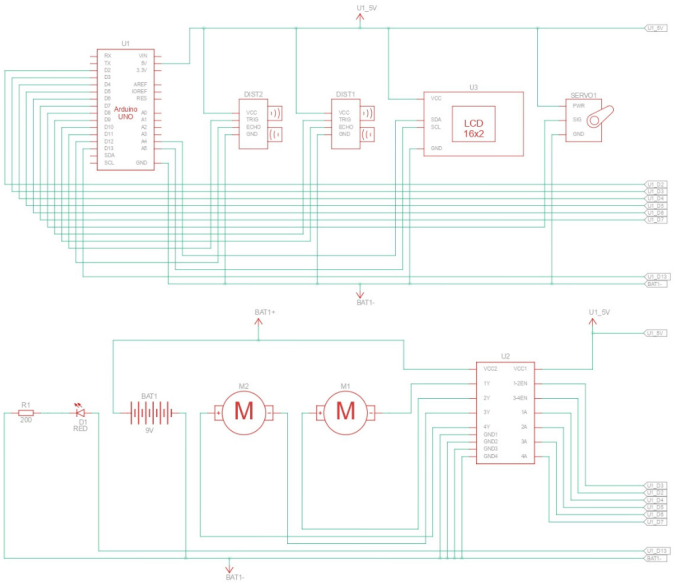

Fig. 5. Circuit Design

V. Simulation Results and Discussion

This work was validated using a simulation-based environment implemented through the dashboard interface. A graph based map was used for representing the navigation space, where start and goal points were defined along with obstacles. The A* pathfinding algorithm was used for calculating the optimal path between nodes. The system also supports dynamic changes in environment by introducing obstacles during navigation. The performance of the system is summarized using different test scenarios categorized based on path length. For each case, parameters such as total distance, computation time, and journey time were recorded. These values help in understanding how the system behaves under different levels of complexity.

TABLE I
Simulation Metrics

Path length	Average Distance (meter)	Average path planning time (ms)	Average journey time (sec)
Very small	110 m	12 ms	244
Small	174 m	15 ms	386
Medium	212 m	19 ms	471
Large	273 m	23 ms	606
Very large	335 m	27 ms	744

The results of the experiments show that as the path grows longer, the calculation time will also become more. This is expected since the bigger the range of search, the greater the number of nodes that will be explored by the algorithm. But despite the complexity of the path, the calculation time remains low enough, indicating the efficiency of the A* algorithm, which utilizes heuristic methods. The importance of the dashboard cannot be overemphasized since it plays a significant role in verifying the effectiveness of the system. It not only shows a graphical depiction of the map and the objects on it but also displays the computed path and the movement of the vehicle. Furthermore, it changes during re-planning, enabling analysis of the behaviour of the system and validation of the results obtained through the implementation of the algorithm.

VI. Future Works

Future work involves taking this proof-of-work framework and implementing it into a real electric vehicle by incorporating the proposed hardware components into it. In addition to that, there could be an incorporation of the perception module with actual sensors rather than simulated inputs. There could be an improvement in the design of the navigation system by adding new and more efficient algorithms for the path planning process or developing hybrid path planning solutions. This would enhance the capability of the system to navigate more challenging environments. Moreover, it would be important to further develop the system to incorporate real-time control capabilities in the system. For example, optimizing its navigation capabilities in terms of speed and efficiency. Such a step is necessary for implementing the system into practical applications. The proposed framework can be expanded to larger-scale environments. For instance, the system can be implemented in industrial settings, warehouses, and smart campuses. The use of IoT and cloud-based monitoring systems can be considered to deploy the proposed solution at large scales. Future works can also improves the safety aspects of the proposed solution, including emergency management, fault detection, and redundancy.

VII. Conclusion

In this paper, an autonomous navigation system that uses a graph-based environment and the A* algorithm has been simulated. The system includes both a dashboard interface to be visualised and monitored and support dynamic re-planning. A hardware design was as well developed and tested with circuit simulation tools. The system can effectively compute

the best paths and react to environmental changes by re-planning. The simulation results reveal that the strategy is effective in navigation tasks in controlled conditions. This paper proposes a solution of low cost and scalability to enable autonomous movement of current vehicles without necessarily having to replace the entire vehicle. The suggested system can find application in places like campuses, industrial premises and other organized places. The system has been tested by simulation by the dashboard and circuit design in TinkerCAD. Such validations reveal that both software and hardware are practical and can be used in the future work. The present implementation however is restricted to simulation based validation and lacks real world hardware validation. The perception module is also simplified and does not use actual sensor data. The system can be further developed in the future with the inclusion of real sensors like cameras and LiDAR, and implementation on a real electric vehicle. Perception and control modules can also be improved to improve performance in real-world situations.

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